

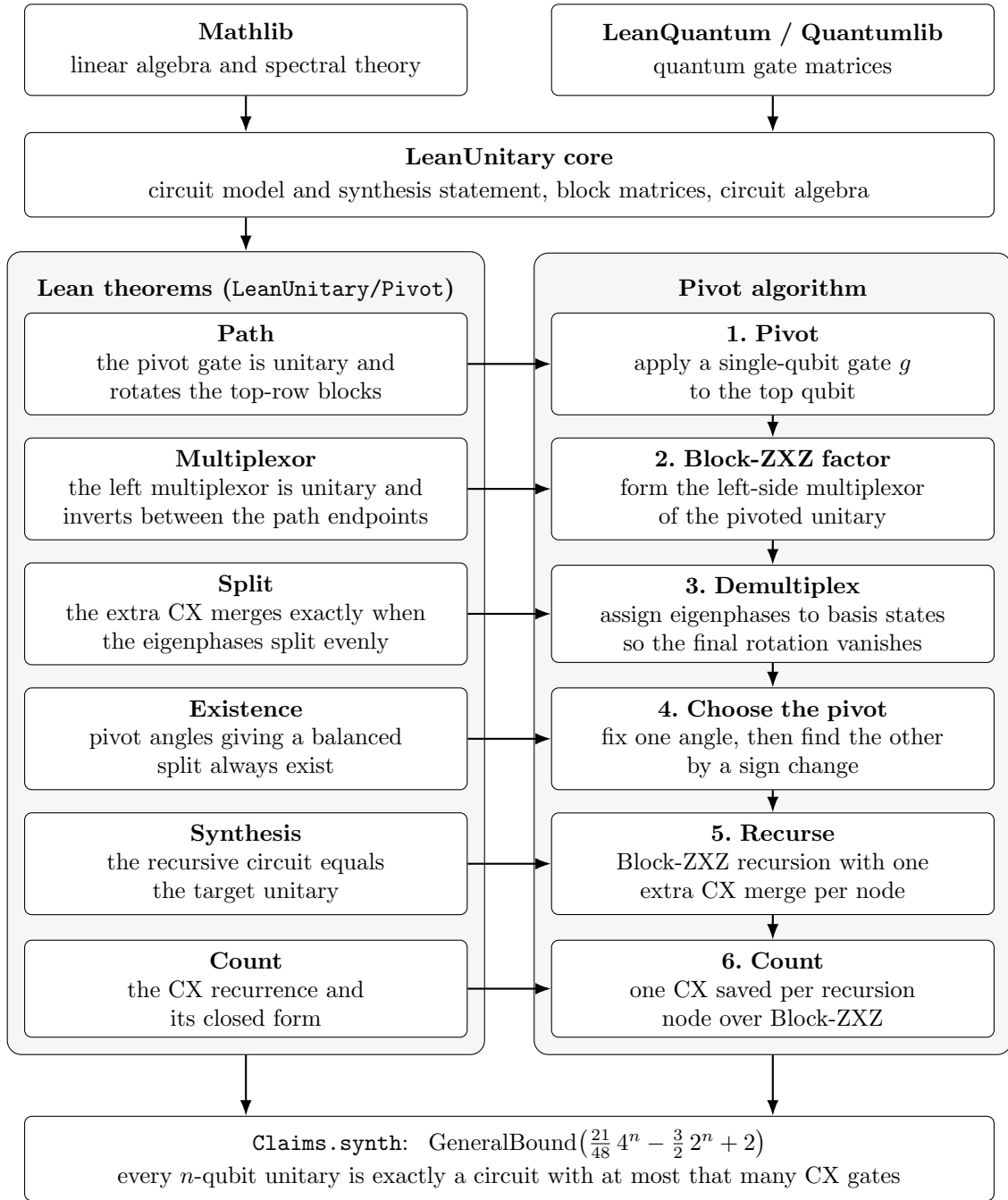


Figure 1: Lean formalization of our proposed pivot decomposition. Each theorem module on the left verifies the step of the pivot algorithm on its right, on top of Mathlib’s linear algebra and the circuit model of LeanUnitary. Together they give the CX bound stated in `Claims.synth`.